

Physics 2 Test 1

Short test reconstructed from the supplied worksheet scans. Show diagrams, sign conventions, intermediate equations, and units. Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ and $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$.

1. Conductors in electrostatics [12 points]

- Explain three characteristics of a conductor in electrostatic equilibrium.
- Draw a spherical conductor in a constant external electric field and show how the field lines bend at the surface.
- Draw the electric-field lines for an irregularly shaped conductor, including the stronger crowding near a sharp point.
- Explain how the result is related to electrostatic shielding.



field lines meet the surface at 90°

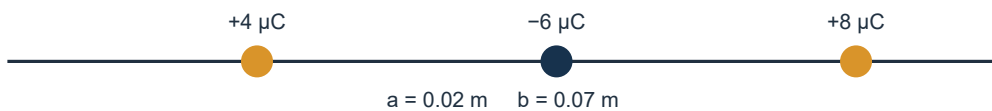


field lines meet the surface at 90°

Electric Fields and Band Diagrams

2. Direct field and energy questions [18 points]

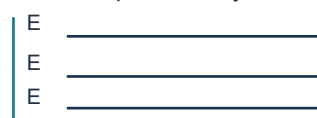
- a. Three point charges lie on a line in the order shown. Calculate the net electric force on the $+8 \mu\text{C}$ charge.



- b. For the rectangular charge arrangement below, give a symbolic solution for the electric field at P, the force on a charge q_4 placed at P, and the electric potential at P. Let $q_1=(0,0)$, $q_2=(0,b)$, $q_3=(a,0)$, and $P=(a,b)$.



- c. Draw the Bohr model of an isolated atom and its corresponding discrete energy-level diagram. Then sketch the energy levels of three identical atoms as their separation decreases, and one periodic crystal model showing valence and conduction bands.

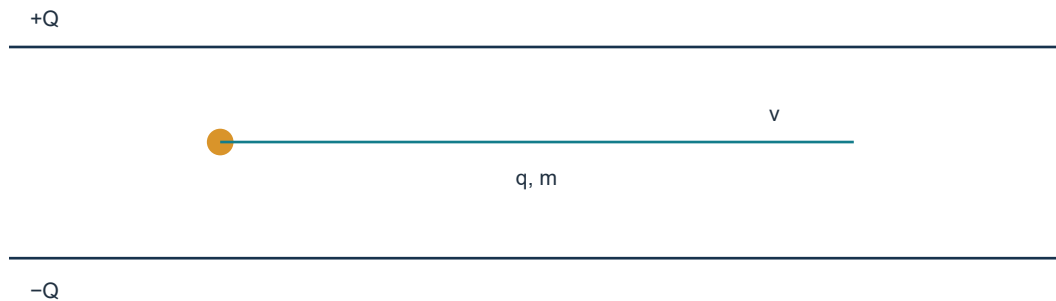


separation $\rightarrow$

Gauss Law and Charge Distributions

3. Symmetry and field derivations [16 points]

- a. A charged particle enters midway between parallel plates of area L^2 . The plates carry $+Q$ and $-Q$. The particle has charge q , mass m , horizontal speed v , and spends time L/v between the plates. Derive the vertical acceleration and vertical deflection.



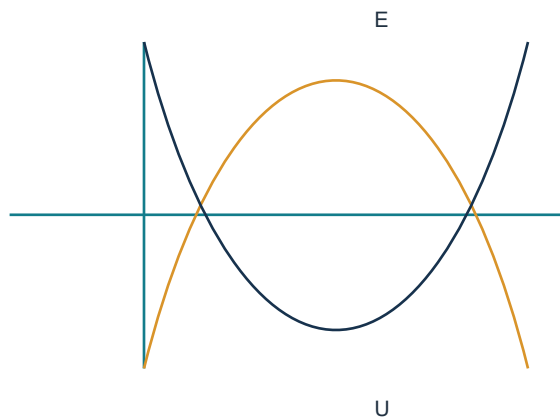
- b. A solid sphere of radius R has volume charge density $\rho(r)=\rho_0(r/R)^2$. Find its total charge by integrating spherical shells.

shell radius r' , thickness dr'

Oscillation and Sound

4. Mass-spring oscillator [12 points]

A 3.3 kg mass is attached to a spring with spring constant 100 N/m. The mass oscillates with amplitude 0.40 m. Calculate: (a) the maximum total mechanical energy, (b) the maximum speed, (c) the speed when $x=0.03$ m, and (d) draw the energy-versus-position curves for kinetic and spring potential energy.



Sound Pressure Level

5. Sound pressure level [10 points]

Define sound-pressure level in terms of rms pressure, using $p_0 = 20 \mu\text{Pa}$. Then determine the change in level when the rms pressure changes from p_1 to $2p_1$ and from p_1 to $\frac{1}{2}p_1$. Explain why doubling pressure does not produce a 2 dB change.

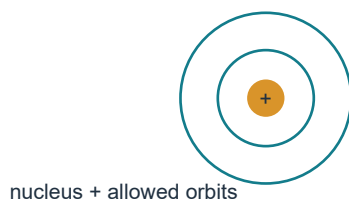
Reminder: intensity for a sinusoidal sound satisfies $I \propto p_{\text{rms}}^2$.



Atomic Models and Molecules

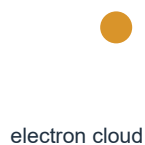
6. Bohr model, ions, and molecules [12 points]

- a. Draw a hydrogen atom using the Bohr picture and a corresponding discrete energy-level diagram. Write the force balance used for a circular electron orbit.



- b. Explain what changes when an atom is ionized. Predict the common ions formed by sodium and chlorine and justify each using valence electrons.

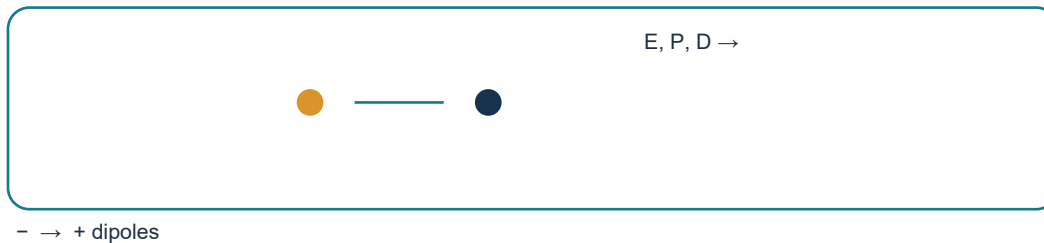
- c. Replace the orbit picture with an electron-cloud picture. Draw one diatomic molecule and one triatomic molecule, labeling the atoms represented.



Dielectrics and Displacement

7. Polarization and displacement vector [10 points]

A dielectric of volume V contains microscopic dipoles $\mathbf{p}_i$. (a) Define the dipole moment and polarization. (b) Starting from $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ and $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, derive $\mathbf{D} = \epsilon \mathbf{E}$ for a linear isotropic dielectric. (c) Draw the relative directions of $\mathbf{E}$, $\mathbf{P}$, and $\mathbf{D}$ and show aligned molecular dipoles.



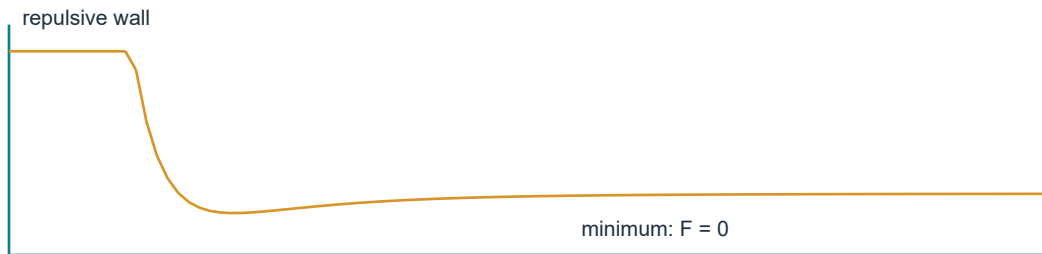
Use standard SI units and identify which quantities describe free charge versus material response.

$$\mathbf{p} = q\mathbf{d} ; \mathbf{P} = (1/V)\sum \mathbf{p}_i ; \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} ; \mathbf{P} = \epsilon_0 \chi_e \mathbf{E} ; \mathbf{D} = \epsilon \mathbf{E}$$

Interaction Potentials

8. Ions and molecular potential energy [12 points]

For two interacting particles: (a) state the point-charge potential energy and the force–potential relation; (b) sketch and label $U(r)$ for a realistic ionic or molecular interaction, including repulsive, attractive, equilibrium, and large- r regions; (c) write the Lennard–Jones form and explain why Na^+ and Cl^- are composite particles rather than literal point charges.



Remember: the force is determined by the slope, not directly by whether U is positive or negative.

$$U = kq_1q_2/r ; F(r) = -dU/dr ; U_{\text{LJ}} = C_{12}/r^{12} - C_6/r^6$$